

Decision Table vs Decision Tree

1. Decision Table

Meaning: A decision table is like a structured table (rows and columns) that lists conditions and possible actions.

It is mainly used when you have many rules and you want to see all conditions in one place.

How it works:

- Conditions are written in columns.
- Possible actions are written as outcomes.
- Each row = one rule.

2. Decision Tree

Meaning: A decision tree is like a flowchart shaped like a tree.

Each node = a decision or condition.

Branches = possible outcomes.

Leaves (end points) = final decision.

How it works:

You start from the root (main condition) and follow branches step by step until you reach a decision.

Example of Decision Table (Loan Approval System):

Condition 1: Income > ■50,000	Condition 2: Credit Score > 700	Decision (Action)
Yes	Yes	Approve Loan
Yes	No	Review Manually
No	Yes	Review Manually
No	No	Reject Loan

Example of Decision Tree (Loan Approval System):

```
Start
|
|-- Is Income > ■50,000?
|   |
|   |-- Yes --> Is Credit Score > 700?
|   |   |
|   |   |-- Yes --> Approve Loan
|   |   |-- No --> Review Manually
|   |
|   |-- No --> Is Credit Score > 700?
|   |   |
|   |   |-- Yes --> Review Manually
|   |   |-- No --> Reject Loan
```

Key Difference

Aspect	Decision Table	Decision Tree
Format	Tabular (rows & columns)	Graphical (tree-like flowchart)

Best For	Many conditions and rules	Step-by-step decision making
Easy to See	All rules at once	Flow of decision process
Example Use	Policy rules, medical diagnosis	Credit scoring, troubleshooting

Summary in one line:

Decision Table = Good for listing all possible condition-action rules in one view.

Decision Tree = Good for showing the step-by-step process of reaching a decision.